Exp. no: 4
07.08.26

Setting up a Local Area Network

Aim:

To design, configure and verify a 4-PC Local Area Network (LAN) using a central Switch, static IP addressing, ping connectivity tests and HTTP resource sharing

Equipment Required:

Equipment	Quantity	Specification
Personal Computer	4	Ethernet (NIC)
Network Switch	1	8/16/24 Port
Ethernet Cables	4	Straight-through

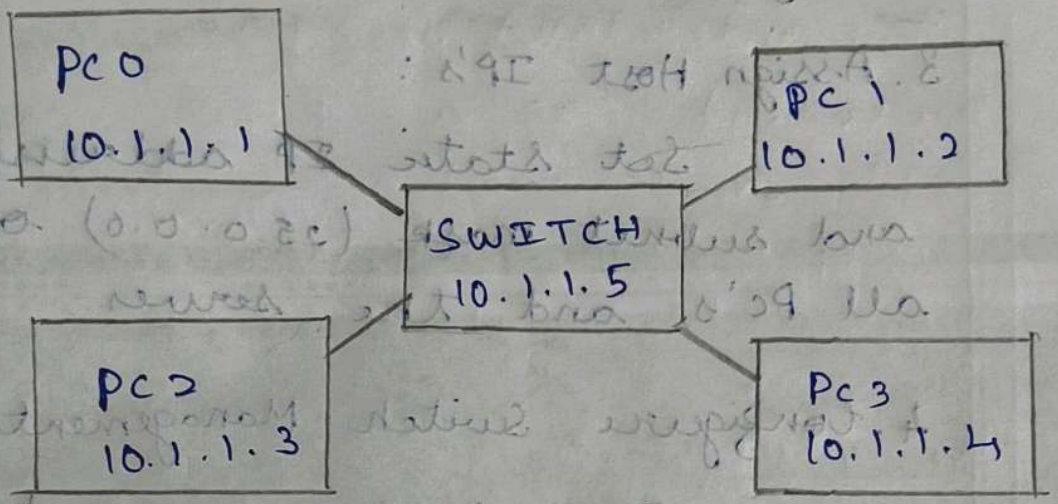
Theory:

A Local Area Network (LAN) connects devices within a limited area to share resources.

This experiment uses a Star Topology where hot connect to a central layer 2 Switch.

Network Diagram

Star Topology LAN



IP Configuration Table:

Device	IP Address	Subnet Mask	Connected
PC 0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC 1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC 2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC 3	10.1.1.4	255.0.0.0	Switch Port Fa 0/4
Switch	10.1.1.5	255.0.0.0	VLAN

Implementation Steps

1. Place Devices
   Drag 4 PC, 1 Server and 1 Switch (2960) to the workspace
2. Cable connections:
   Connect PC's (Fa 0/1 to Fa 0/4)

and server (Fa 0/5) to switch using copper straight cables

3. Assign Host IP's:

Set Static IP addresses and subnet mask (25.0.0.0) on all Pc's and the server.

4. Configure Switch Management:

Open Switch CLI and run enable.

configure terminal
interface Vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit

5. Verify Connectivity

Run Ping 10.1.1.2 from PC0 command prompt to verify network link

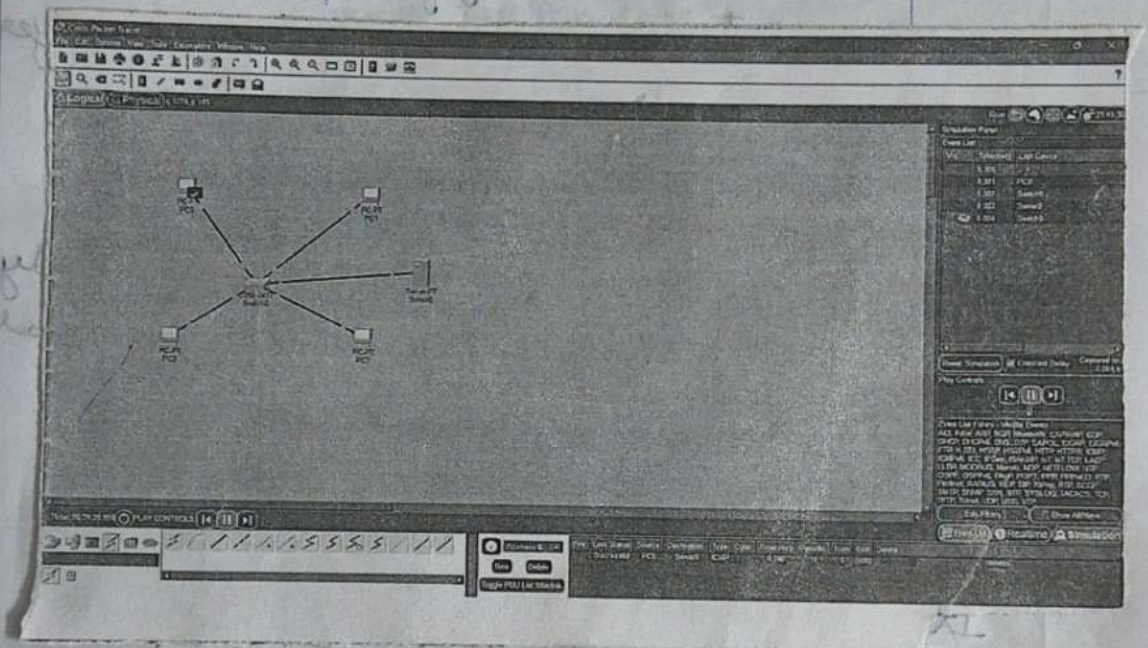
6. Demonstrate Resource sharing

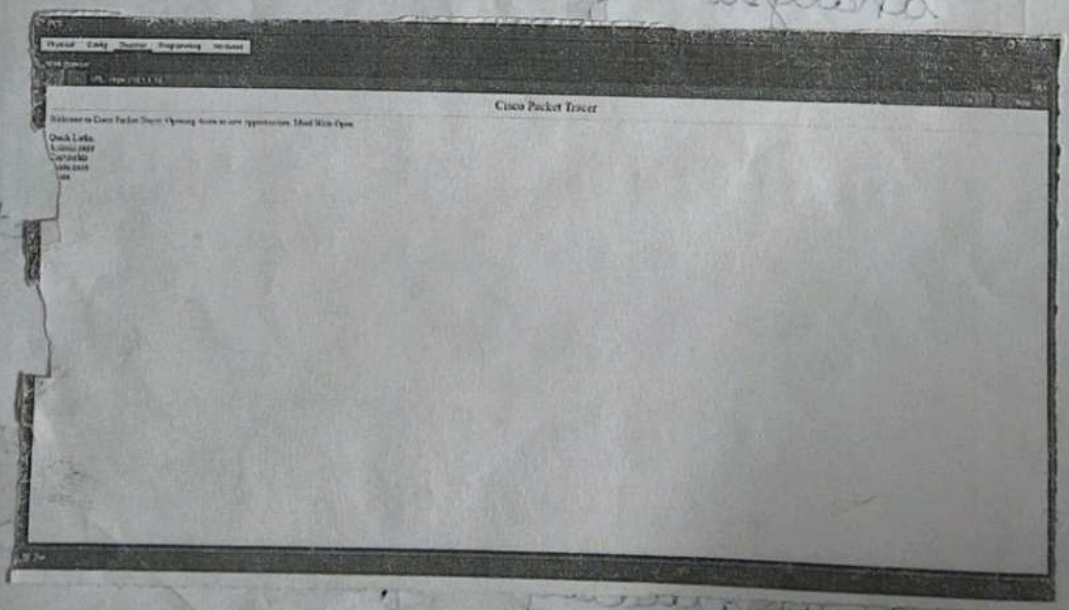
Enable HTTP on Server (10.1.1.10) open client PC web browser and navigate to [http://10.1.1.10]

7. Observe Packet flow :

Switch to simulation mode to trace ARP and ICMP/HTTP frame forwarding across ports

Output





Result :

The LAN lab experiment was successfully built and verified for IP connectivity.